

HDOC Day-11 Activity

Menu-Driven C Program using Switch-Case

1. Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Find the sum of squares of its digits using a while loop.
- Find the sum of cubes of its digits using a for loop.
- Find the difference between the sum of cubes and the sum of squares of its digits using a do-while loop.

2. Algorithm

1. Start the program.
2. Declare variables: num, n, digit, choice, sumSquares, sumCubes, diff.
3. Read a positive integer 'num' from the user.
4. Display a menu with three options and read the user's 'choice'.
5. Using switch-case, evaluate 'choice':
 - Case 1: Use a while loop to extract each digit of num, square it, and add it to sumSquares. Print sumSquares.
 - Case 2: Use a for loop to extract each digit of num, cube it, and add it to sumCubes. Print sumCubes.
 - Case 3: Use a do-while loop to extract each digit of num, accumulating both sumCubes and sumSquares, then compute $\text{diff} = \text{sumCubes} - \text{sumSquares}$. Print diff.
6. For an invalid choice, display an appropriate message (default case).
7. Stop the program.

3. Test Case Table

Choice	Input Number	Expected Output	Actual Output	Result
1	123	14	Sum of squares of digits = 14	Pass
2	123	36	Sum of cubes of digits = 36	Pass
3	123	22	Difference between sum of cubes and sum of squares = 22	Pass
1	0	0	Sum of squares of digits = 0	Pass
2	9	729	Sum of cubes of digits = 729	Pass
3	999	1944	Difference between sum of cubes and sum of squares = 1944	Pass

4. C Program

```
#include <stdio.h>

int main() {
```

```

int num, n, digit, choice;
long sumSquares = 0, sumCubes = 0, diff = 0;

printf("Enter a positive integer: ");
scanf("%d", &num);

printf("Menu:\n");
printf("1. Sum of squares of digits (while loop)\n");
printf("2. Sum of cubes of digits (for loop)\n");
printf("3. Difference between sum of cubes and sum of squares (do-while loop)\n");
printf("Enter your choice: ");
scanf("%d", &choice);

switch (choice) {
    case 1:
        n = num;
        while (n != 0) {
            digit = n % 10;
            sumSquares += digit * digit;
            n = n / 10;
        }
        printf("Sum of squares of digits = %ld\n", sumSquares);
        break;

    case 2:
        for (n = num; n != 0; n = n / 10) {
            digit = n % 10;
            sumCubes += digit * digit * digit;
        }
        printf("Sum of cubes of digits = %ld\n", sumCubes);
        break;

    case 3:
        n = num;
        do {
            digit = n % 10;
            sumCubes += digit * digit * digit;
            sumSquares += digit * digit;
            n = n / 10;
        } while (n != 0);
        diff = sumCubes - sumSquares;
        printf("Difference between sum of cubes and sum of squares = %ld\n", diff);
        break;

    default:
        printf("Invalid choice!\n");
}

return 0;
}

```

5. Compilation and Execution

The program was compiled and executed on a Linux terminal using the GCC compiler as follows:

```
$ cc hdoc11.c a.out
```

```
Session Actions Edit View Help

(kali@kali)-[~]
$ nano hdoc11.c

(kali@kali)-[~]
$ cc hdoc11.c

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 123
Menu:
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice: 1
Sum of squares of digits = 14

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 123
Menu:
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice: 2
Sum of cubes of digits = 36

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 123
Menu:
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice: 3
Difference between sum of cubes and sum of squares = 22

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 0
Menu:
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice: 1
Sum of squares of digits = 0

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 9
Menu:
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice: 2
Sum of cubes of digits = 729

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 999
Menu:
1. Sum of squares of digits (while loop)
2. Sum of cubes of digits (for loop)
3. Difference between sum of cubes and sum of squares (do-while loop)
Enter your choice: 3
Difference between sum of cubes and sum of squares = 1944
```

6. Conclusion

The menu-driven C program was successfully written, compiled, and executed using switch-case along with while, for, and do-while loops. All test cases produced the expected output, verifying the correctness of the program.